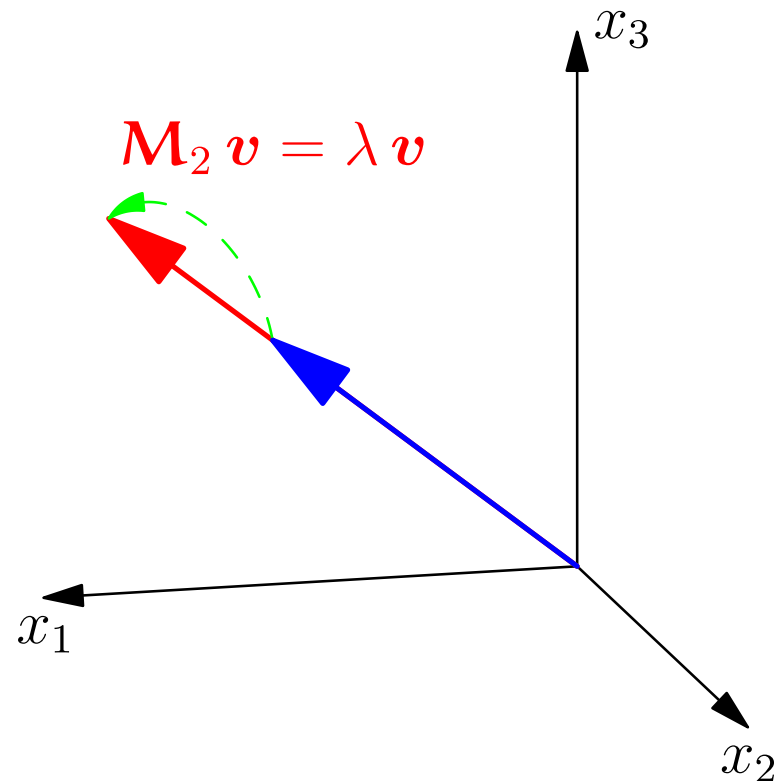


Mathematics of Machine Learning

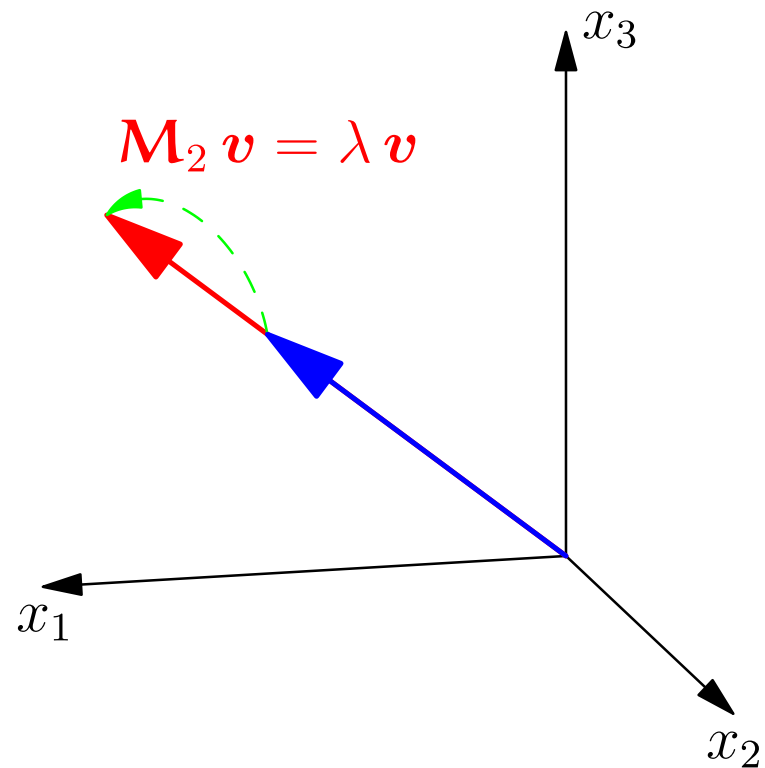
Eigensystems



Eigenvectors, Orthogonal Matrices, Eigenvector Decomposition, Rank

Outline

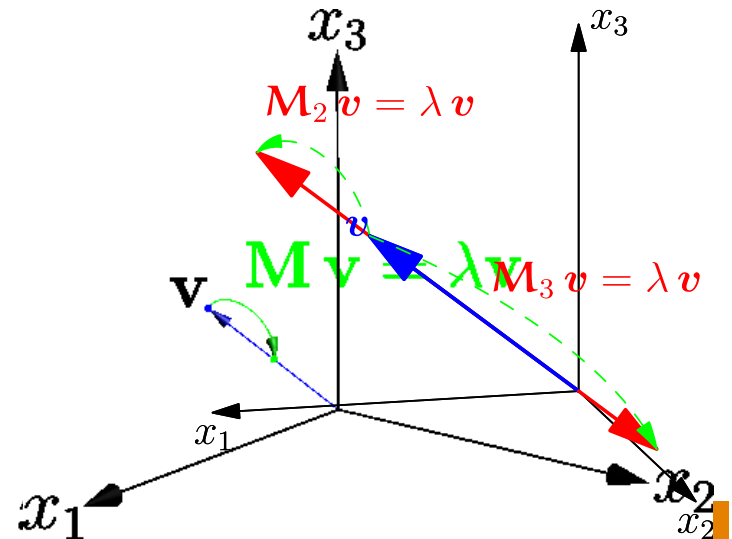
1. **Eigenvectors**
2. Orthogonal Matrices
3. Eigen Decomposition
4. Low Rank Approximation



Eigenvector equation

- Eigen-systems help us to understand mappings
- A vector v is said to be an **eigenvector** if

$$Mv = \lambda v$$



- M is square (i.e. $n \times n$)
- Where the number λ is the **eigenvalue**
- Eigenvalues play a fundamental role in understanding operators

Symmetric Matrices

- If $\mathbf{M}$ is an $n \times n$ **symmetric** matrix then it has n real orthogonal eigenvectors with real eigenvalues■
- We denote the i^{th} eigenvector by $\mathbf{v}_i$ and the corresponding eigenvalue by λ_i so that

$$\mathbf{M}\mathbf{v}_i = \lambda_i \mathbf{v}_i \blacksquare$$

- Orthogonal means that if $i \neq j$ then

$$\mathbf{v}_i^T \mathbf{v}_j = 0 \blacksquare$$

- (We can always normalise eigenvectors if we want)■

Proof of Orthogonality

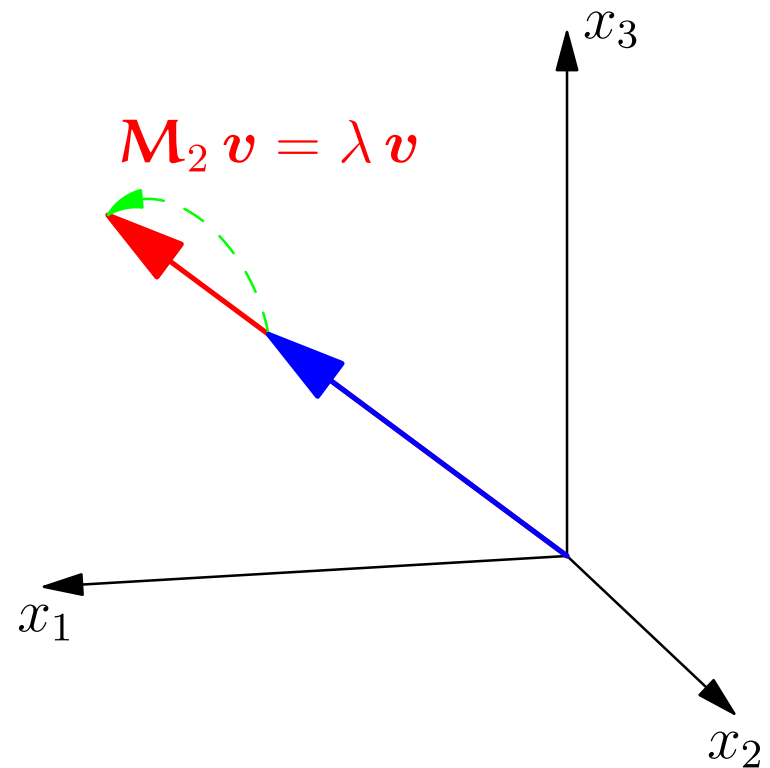
- $(\mathbf{M}\mathbf{v}_i = \lambda_i\mathbf{v}_i)^\top$ implies $\mathbf{v}_i^\top \mathbf{M}^\top = \lambda_i\mathbf{v}_i^\top$ ■
- When $\mathbf{M}$ is symmetric then $\mathbf{M}\mathbf{v}_i = \lambda_i\mathbf{v}_i \Rightarrow \mathbf{v}_i^\top \mathbf{M} = \lambda_i\mathbf{v}_i^\top$ ■
- Consider two eigenvectors $\mathbf{v}_i$ and $\mathbf{v}_j$ of $\mathbf{M}$

$$\begin{aligned}\mathbf{v}_i^\top \mathbf{M}\mathbf{v}_j &= (\mathbf{v}_i^\top \mathbf{M})\mathbf{v}_j = \lambda_i\mathbf{v}_i^\top \mathbf{v}_j \\ &= \mathbf{v}_i^\top (\mathbf{M}\mathbf{v}_j) = \lambda_j\mathbf{v}_i^\top \mathbf{v}_j\end{aligned}$$

- So either $\lambda_i = \lambda_j$ or $\mathbf{v}_i^\top \mathbf{v}_j = 0$ ■
- If $\lambda_i = \lambda_j$ then any linear combination of $\mathbf{v}_i$ and $\mathbf{v}_j$ is an eigenvector $(\mathbf{M}(a\mathbf{v}_i + b\mathbf{v}_j) = \lambda_i(a\mathbf{v}_i + b\mathbf{v}_j))$ ■ So I can choose two eigenvectors that are orthogonal to each other. ■

Outline

1. Eigenvectors
2. **Orthogonal Matrices**
3. Eigen Decomposition
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Orthogonal Matrices

- We can construct an **orthogonal** matrix $\mathbf{V}$ from the eigenvectors

$$\mathbf{V} = (\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n)$$

- Matrix $\mathbf{V}$ is an $n \times n$ matrix
- Because of the orthogonality of the vectors $\mathbf{v}_i$

$$\mathbf{V}^T \mathbf{V} = \begin{pmatrix} \mathbf{v}_1^T \mathbf{v}_1 & \mathbf{v}_1^T \mathbf{v}_2 & \cdots & \mathbf{v}_1^T \mathbf{v}_n \\ \mathbf{v}_2^T \mathbf{v}_1 & \mathbf{v}_2^T \mathbf{v}_2 & \cdots & \mathbf{v}_2^T \mathbf{v}_n \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{v}_n^T \mathbf{v}_1 & \mathbf{v}_n^T \mathbf{v}_2 & \cdots & \mathbf{v}_n^T \mathbf{v}_n \end{pmatrix} = \begin{pmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{pmatrix} = \mathbf{I}$$

The Other Way Around

- We have shown that $\mathbf{V}^T \mathbf{V} = \mathbf{I}$
- Thus multiply both sides on the left by $\mathbf{V}$

$$\mathbf{V} \mathbf{V}^T \mathbf{V} = \mathbf{V}$$

- $\mathbf{V}$ will have an inverse, $\mathbf{V}^{-1}$, such that $\mathbf{V} \mathbf{V}^{-1} = \mathbf{I}$
- Multiplying the equation on the right by $\mathbf{V}^{-1}$

$$(\mathbf{V} \mathbf{V}^T) \mathbf{V} \mathbf{V}^{-1} = \mathbf{V} \mathbf{V}^{-1}$$

$$\mathbf{V} \mathbf{V}^T = \mathbf{I}$$

- Note that, $\mathbf{V}^{-1} = \mathbf{V}^T$ (definition of orthogonal matrix)

Invertible Matrices

- A matrix, $\mathbf{M}$, will be singular (uninvertible) if there exists a vector $\mathbf{x} (\neq \mathbf{0})$ such that

$$\mathbf{M}\mathbf{x} = \mathbf{0}$$

- Now if there exists such a vector such that $\mathbf{V}\mathbf{x} = \mathbf{0}$ then multiply by $\mathbf{V}^T$ we get

$$\mathbf{V}^T\mathbf{V}\mathbf{x} = \mathbf{V}^T\mathbf{0}$$

$$\mathbf{x} = \mathbf{0}$$

since $\mathbf{V}^T\mathbf{V} = \mathbf{I}$

- Thus $\mathbf{V}$ is invertible

Rotations

- Orthogonal matrices satisfy $\mathbf{V}^T \mathbf{V} = \mathbf{V} \mathbf{V}^T = \mathbf{I}$
- As a consequent they define rotations (and possibly a reflection)
- Consider a vector x and $x' = \mathbf{V}x$, now

$$\|x'\|_2^2 = x'^T x' = (\mathbf{V}x)^T (\mathbf{V}x) = x^T \mathbf{V}^T \mathbf{V} x = x^T x = \|x\|_2^2$$

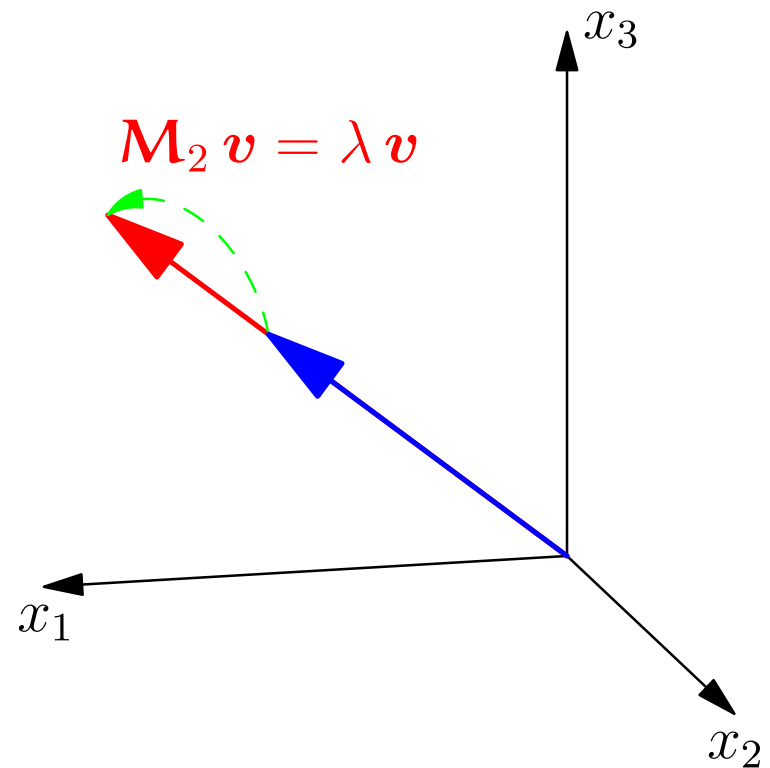
- Similarly if additionally $y' = \mathbf{V}y$ then

$$\langle x', y' \rangle = (\mathbf{V}x)^T (\mathbf{V}y) = x^T \mathbf{V}^T \mathbf{V} y = x^T y = \langle x, y \rangle = \|x\|_2 \|y\|_2 \cos(\theta)$$

- Rotations and reflections preserve lengths and angles

Outline

1. Eigenvectors
2. Orthogonal Matrices
3. **Eigen Decomposition**
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Matrix Decomposition

- Taking the matrix of eigenvectors, V , then

$$MV = M(v_1, v_2, \dots, v_n) = (\lambda_1 v_1, \lambda_2 v_2, \dots, \lambda_n v_n) = V\Lambda$$

- where $\Lambda = \text{diag}(\lambda_1, \lambda_2, \dots, \lambda_n) = \begin{pmatrix} \lambda_1 & 0 & \dots & 0 \\ 0 & \lambda_2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \lambda_n \end{pmatrix}$

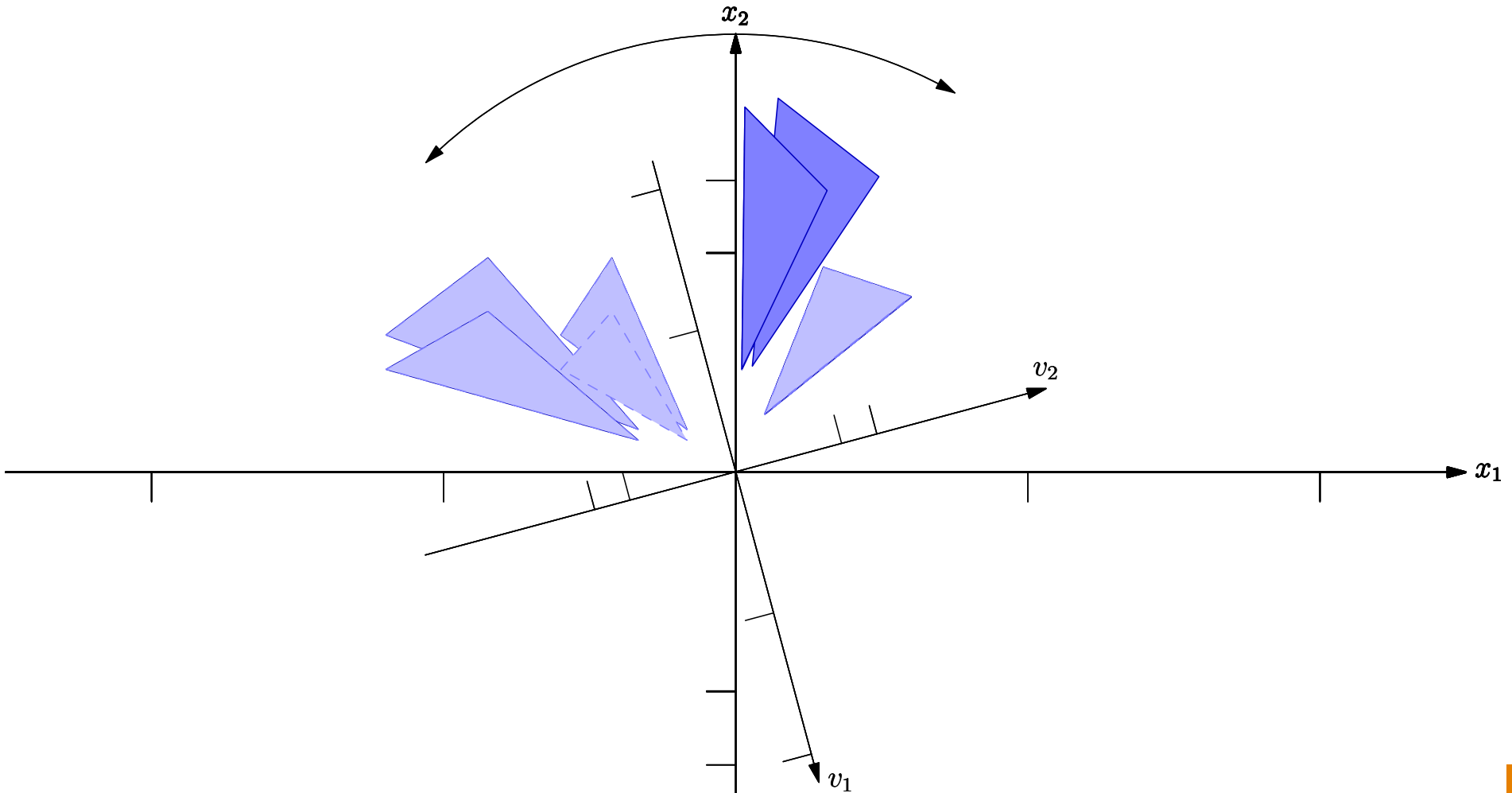
- Now

$$M = MVV^T = V\Lambda V^T$$

- Very important *similarity transform*

Mappings by Symmetric Matrices

$$\mathbf{M} = \begin{pmatrix} 0.83 & -0.31 \\ -0.31 & 1.9 \end{pmatrix} = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^T = \begin{pmatrix} \cos(-75) & \sin(-75) \\ -\sin(-75) & \cos(-75) \end{pmatrix} \begin{pmatrix} 2 & 0 \\ 0 & 0.75 \end{pmatrix} \begin{pmatrix} \cos(75) & \sin(75) \\ -\sin(75) & \cos(75) \end{pmatrix}$$



Inverses

- For any symmetric invertible matrix

$$\mathbf{M} = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^T$$

$$\mathbf{M}^{-1} = \mathbf{V}\mathbf{\Lambda}^{-1}\mathbf{V}^T$$

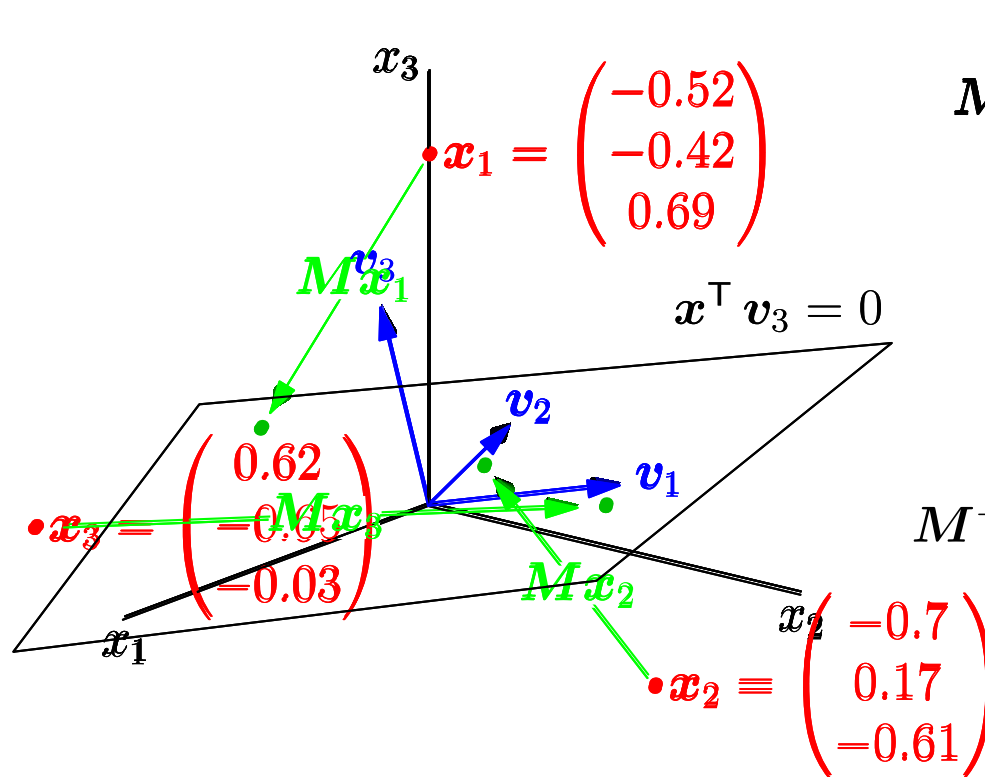
- Where $\mathbf{\Lambda}^{-1} = \text{diag}(\frac{1}{\lambda_1}, \frac{1}{\lambda_2}, \dots, \frac{1}{\lambda_n}) = \begin{pmatrix} \frac{1}{\lambda_1} & 0 & \dots & 0 \\ 0 & \frac{1}{\lambda_2} & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \frac{1}{\lambda_n} \end{pmatrix}$

- Since

$$\begin{aligned} \mathbf{M}\mathbf{M}^{-1} &= (\mathbf{V}\mathbf{\Lambda}\mathbf{V}^T)(\mathbf{V}\mathbf{\Lambda}^{-1}\mathbf{V}^T) = \mathbf{V}\mathbf{\Lambda}(\mathbf{V}^T\mathbf{V})\mathbf{\Lambda}^{-1}\mathbf{V}^T \\ &= \mathbf{V}\mathbf{\Lambda}\mathbf{\Lambda}^{-1}\mathbf{V}^T = \mathbf{V}\mathbf{V}^T = \mathbf{I} \end{aligned}$$

- I.e, Small eigenvalues become large eigenvalues and visa versa

III-Conditioning Again



$$M = \begin{pmatrix} 0.80 & 0.01 & 0.25 \\ 0.01 & 0.85 & 0.48 \\ 0.25 & 0.48 & 0.5 \end{pmatrix}$$

$$= V \begin{pmatrix} 0.8 & 0 & 0 \\ 0 & 1.25 & 0 \\ 0 & 0 & 0.1 \end{pmatrix} V^T$$

$$M^{-1} = \begin{pmatrix} 1.9 & 1.17 & -2.1 \\ 1.17 & 3.34 & -3.8 \\ -2.1 & -3.8 & 6.8 \end{pmatrix}$$

$$= V \begin{pmatrix} 1.25 & 0 & 0 \\ 0 & 0.8 & 0 \\ 0 & 0 & 10 \end{pmatrix} V^T$$

Condition Number

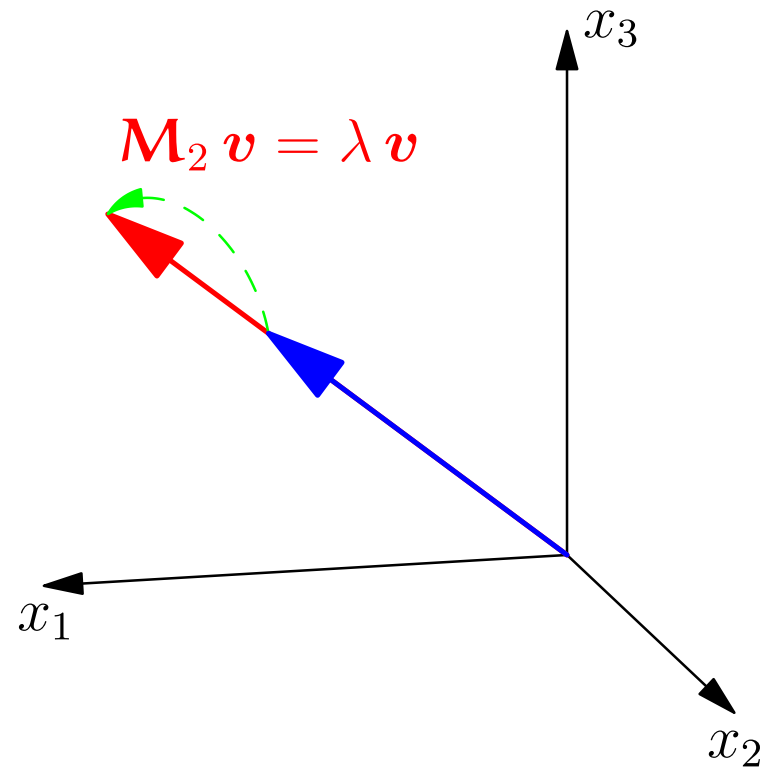
- Taking matrix inverses can be inherently unstable■
- Any small error can be amplified by taking the inverse■
- The stability of the inverse depends on the ratio of smallest eigenvalue to the largest eigenvalue (i.e. the biggest possible amplification compared to the smallest)■
- For invertible matrices we can take the largest eigenvalue as a norm of the matrix■
- The condition number is given by

$$\|\mathbf{M}\|_H \times \|\mathbf{M}^{-1}\|_H = \frac{|\lambda_{\max}|}{|\lambda_{\min}|} \blacksquare$$

- Large condition number implies very ill-conditioned■

Outline

1. Eigenvectors
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Rank of a Matrix

- The rank of a matrix, $\mathbf{M}$, is the number of non-zero eigenvalues■
- The space spanned by the eigenvectors $\mathbf{v}_a$, $\mathbf{v}_b$, etc. with zero eigenvalue forms a **null space**■
- Any vector in the null space will get projected to the zero vector

$$\mathbf{M}(a\mathbf{v}_a + b\mathbf{v}_b + \dots) = \mathbf{0}■$$

- A square matrix is said to be **rank deficient** if it has any eigenvectors with eigenvalue equal to 0■
- This happens when the columns of the matrix are not linearly independent■

“Inverting” Rank Deficient Matrices

- Rank deficient matrices are non-invertible (i.e. we don't know the vector x such that $Mx = b$) as we don't know the component of the x in the null space■
- Although we don't know x we can find a vector, x , that satisfies $Mx = b$ ■
- Given a symmetric $n \times n$ matrix with k non-zero eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_k$ we can construct a “pseudo inverse” M^+ as $V\Lambda^+V^T$ where $\Lambda^+ = \text{diag}(\frac{1}{\lambda_1}, \frac{1}{\lambda_2}, \dots, \frac{1}{\lambda_k}, 0, \dots, 0)$ ■
- This finds the vector x with no component in the null space■(it is the solution with the smallest norm)■
- This is different to the pseudo inverse for non-square matrices■

Low Rank Approximation

- Recall that matrices with large and small eigenvalues are ill-conditioned so the inverse has the potential to greatly amplify any measurement error■
- One work around is to set all small eigenvalues to zero and use the pseudo inverse■
- Setting small eigenvalues to zero reduces the rank of the matrix and is an example of a low rank approximation■
- Low rank approximations are much used to obtain approximate models for arrays of data■ (we will revisit this when we look at SVD)■

Summary

- Linear mappings are commonly used in machine learning algorithms such as regression■
- We can understand symmetric operators by looking at their eigenvectors■
- Any symmetric matrix can be decomposed as $\mathbf{M} = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^T$
 - ★ where $\mathbf{V}$ are orthogonal matrices whose rows are the eigenvector
 - ★ and $\mathbf{\Lambda}$ is a diagonal matrix of the eigenvalues■
- This decomposition allows us to understand inverse mappings■